days.

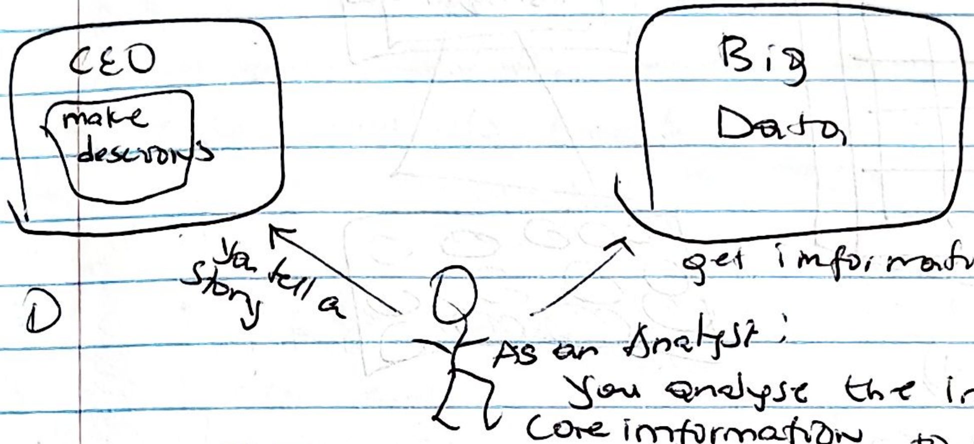
Data Visualisation: Data Story telling

Data: is information that need to make sence for analotics to narrate.

-> Need to be able to tell a story from your data.

-> You need to drive | Extract actionable insights from your data, Source

-> Drive Business descions.



get information from the Source

As an Analyst:
You analyse the insights
Core information to make decisions

Data Process: (RDB) (Excel)

In Relational database: This is where we use Structured data (facts) and dimension tables that can be connected to one another using Primary key & Foreign key.

-> We store structured data has Fact and dimension purpose that canbe connected 2gda using 2 keys.

Primary key is a Unique identify that diffrenciates each componante from the Customers table e.g. Id number: name : Age.

- Contain descriptive information
- A Dim table does not considerly Changes (Id-name-Location, Age)
- Dimensional tables Contain Primary keys
- Dimensional tables

Customer table

Dimensional data

Customer Id.	Name	Age.	City
1	Nthabi	33	~~K~~ZN
2.	JEFF	22	LA
3.	Mark	16	MG

Production - DIM TABLE

Production ID	Production Category	Producti NAME
101	Fruits	Bannena
102	Fruits	apples
103	Electronics	Laptop
(104)	Clothing	Uzzi Jeans

etc,

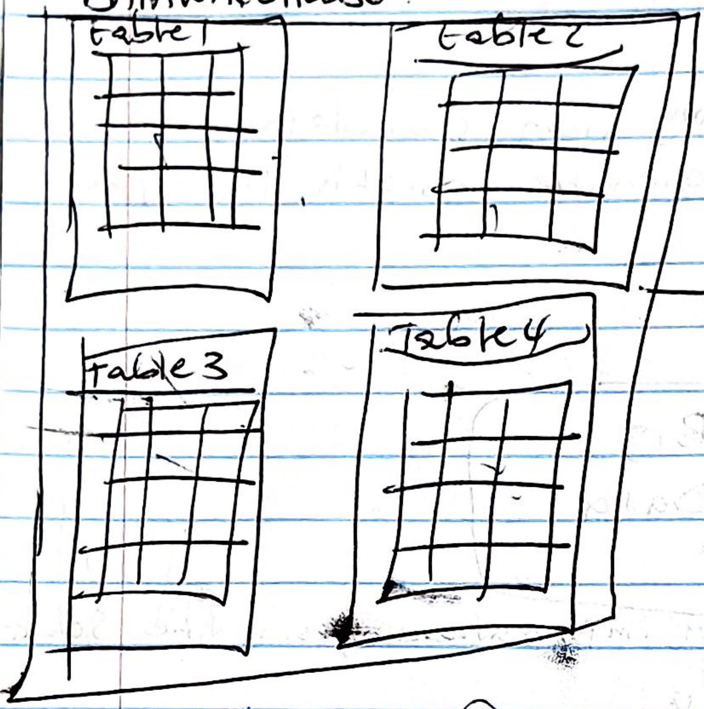
- Transactions - (FACTUAL data)

order-ID.	Customer Id,	Product ID	. Revenue
324	2	(104)	70
333	1	102	80
401	3	101	100

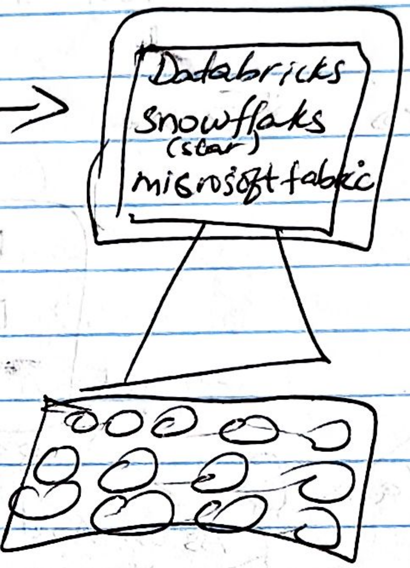
- This are Fact tables (Numerical data)
- they Foreign keys
- They contain numbers
- 104 (Foreign key was sold to Custome 2. Which is Jeff.
- Used For analysis.
- Its a duplicate key.
- In Fact tables you audit:

: **Data Process!** This is Where Online Analytic Process happen - via diffrent Pipelines.

DATAWAREHOUSE :



-> We use DMS to acess data
- Connected to our data.
- in DMS: thats where we CODE



②

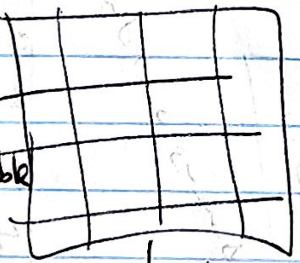
WHY DO WE NEED TO CODE!

- To retrieve, - to amend, to
* update, to manupulate
- to clean, to aggregate
- to delete, to join and also to keep up update w the goal of extracting actionable insights. - Compress for Smaller potion to work on it!

: Also to do inspection | Exploration
↳ to see what data is available
: Data Exploratory Analysis
↳ to understand the relationship between our data points.
Feature Buildings!
↳ Creating future prediction in ML (Marchine Learning)

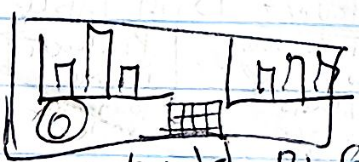
What happens after Coding? and Science You end up

- Summrized table
- Aggregated table
- Joind | combined Clean table
- Unified table
- with rigt features | data.
- has insight table



Thrs helps you to tell a Story that makes sense, from Facts and Dimesions Reporting tool!

↓ then You desicide to build a power report. / Power BI



with all right imfomtion

You build BI from One unified table NOT from warehouse

Claude AI

BAR-CHART! ~~A~~AReea Chat! or of the graph?

• Why pie chart?! Before visualize: - Know what kind of Story you wanna tell.

1. graph
2. How to use
3. When to use
4. Why?

- type o graph
- How to use it.
- When

- Area Chat'

(PART OF 6 day $)

BRS - Business Requirment - (the Ask)

what if need to move forward.

(maybe understand the busines Revenue

→ data Analysts comes

- Reasons for ~~graph~~ buildig a dashbord.

BEFOR REPORTS
1. Understand the Ask
2. Ask Relevant Qs

(in micro (FOR VISUALIZATION))

BRS - Business Requirement. The Ask - The what Reason Behind (The Why?!) ①

Answer

CEO - I want to understand the revenue for our Coffee Shop for different product.

As an Analyst: Before answer the Ask. ASK DATA DRIVEN Questions!

Understand what is needed by asking driven Question (Right)

- which product would you like to see the daily Revenue
- would you like to see the repeat Customer (Returning customers)
- to analyze the frequency of Purchases?

• (What question is it answering).

do you want Revenue with Vat included or Exluded.

• See the frequence of Purchase

daily Revenue?

• Revenue calculations

- data processing
↓
data Analysis (Insight Generation)
↓
Visualization (Creating graphs)
↓
- data Presenting (putting thing 2gether)

code

Code

putting thing 2gether.

(• Understanding the overlapping processing through the systems of presentations)

• Under the Reason behind it.

all these Q are created Before a dashboard; then you can do data processing then do data visualization.

WHY Graphs, boards and Questoons?!!
- Conecti ω Shareholder; understand the consepts

Data Stories
Data Studio
charts caparison

→ FOR DASHBoard to be made.)

① (CEO - wants to track the
Yerenue daily from the Coffee)
by 4 differet productus

② → As for Analyst; ASK releved QS
- data driven QS - (all ansuered by
the presented data @r @ hand).

- use miro to develop
a start to develop Qs.

"Q's" Would You like to see 1 - revenue daily
2 the Report of repeated customers
3 Analys the frequency
4 - Revenue ω tax? or excluded.
(11) - introgate more -
5 - Mesuerments "Year - Year"

③ they will then be gradualize - high should it be?
6 - How the dashboard should like
- What KPIs would you like to
7 ~~measere~~ measure? Revenue? # of
customers? 8 Distonct Products?
Measure Y-Y? M-M? W-W?
Key Perfomee (indicators) You need to see?
MTD - Mont to date - Year to date
T-1 VS T-7 (todays Revenue vs 7day ago

Data Processing:
code
→
Data Analysis:
Insights Generations
Data Visualization:
Greating graphs
↓
Data Presentation:
Putting things together

- Non-Relation DB -
- Organized data is
anything not Relation / traditional
table.
- Key-Value Store
- Documents (xml, JSON)
- Graphs
- Flexible Tables

DAY 6 | Story telling

data Visualization KEY: WHEN and Why Graph to use!

WHEN Dealing ω data / Different Graphing:

different Reasons

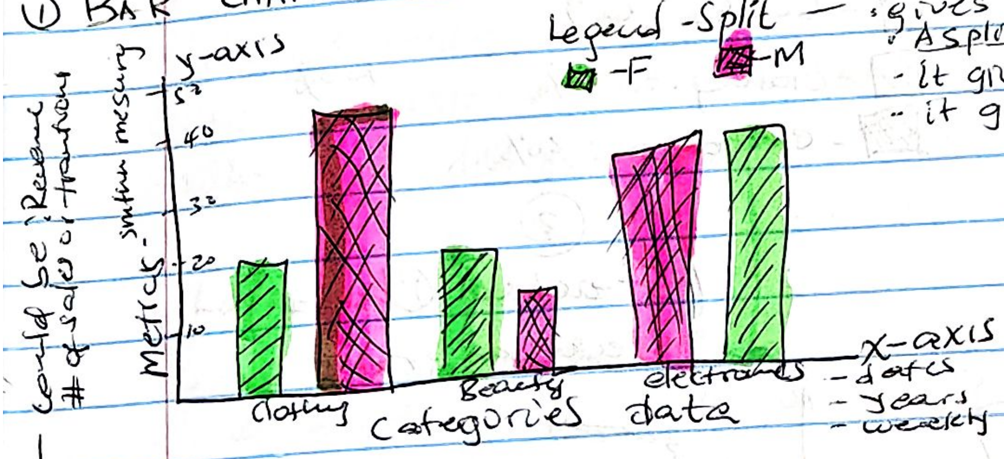
① the Data design blueprint:

(Choosing the Right chart or graph for effective Story telling) ✻ - IN comparison in category

① BAR CHART: When do we use it?

Legend - Split — · categorical data · gives u a breakdown · A split or legend

- It gives you granularity
- it granulize data.



· cycle trends for short actives

↓ # of customers

- its a form of Scale
- (Its a numerical data of Quantitative data)

— Categorical data —

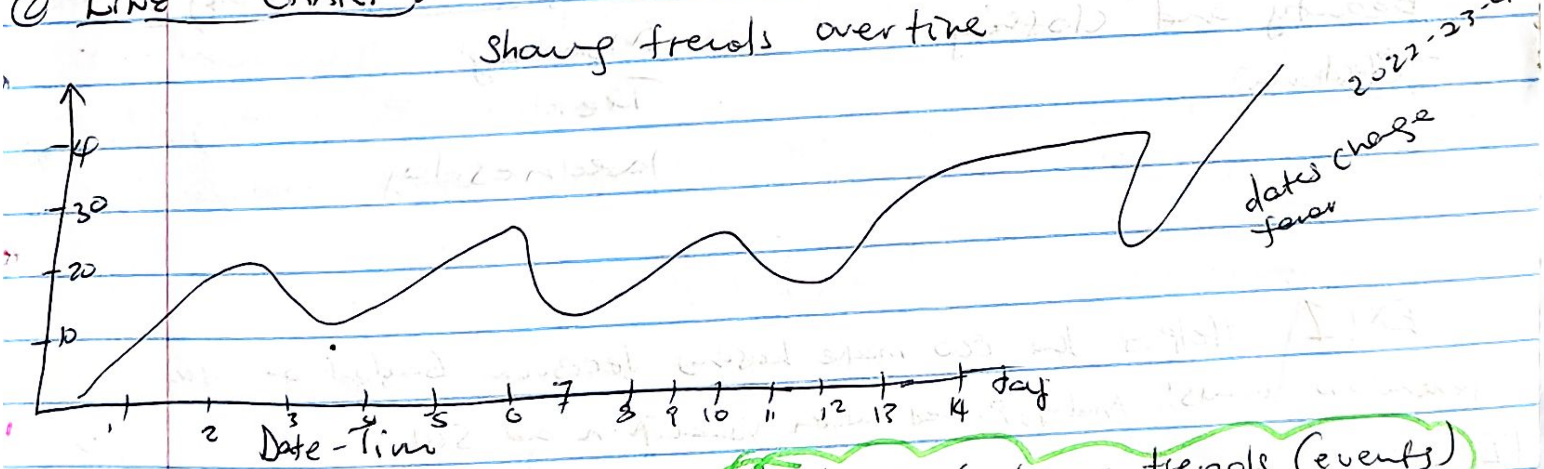
✻ grouping data

✻ Classification data: It puts information into groups: words

eg gender: clothing. beauty. electronics

② LINE CHART:

Showing trends over time

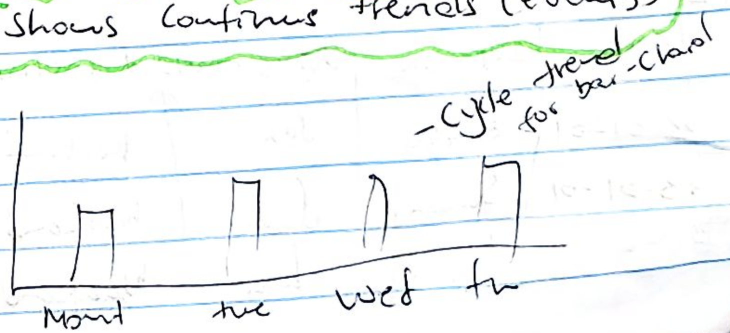


↳ month

↳ day of week

- Year - 22.23.4.25.26
- data
- hours

- It shows continuous trends (events)

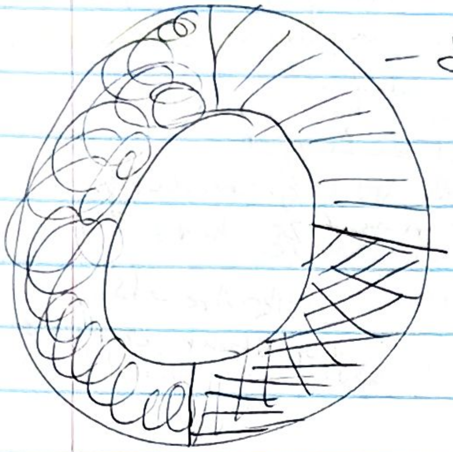


Showing proportions: PIZZA

You need categorical breakdown
- metric to propotion

- what distribution give more revenue?
- understand what customers want.

①



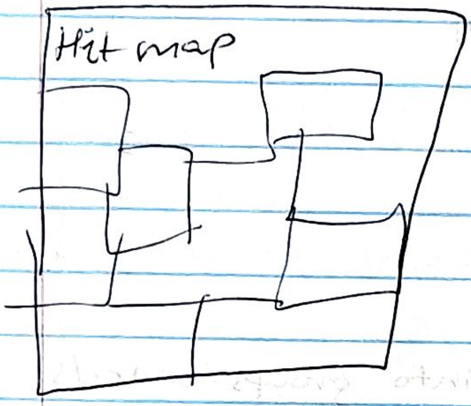
- dounught chart

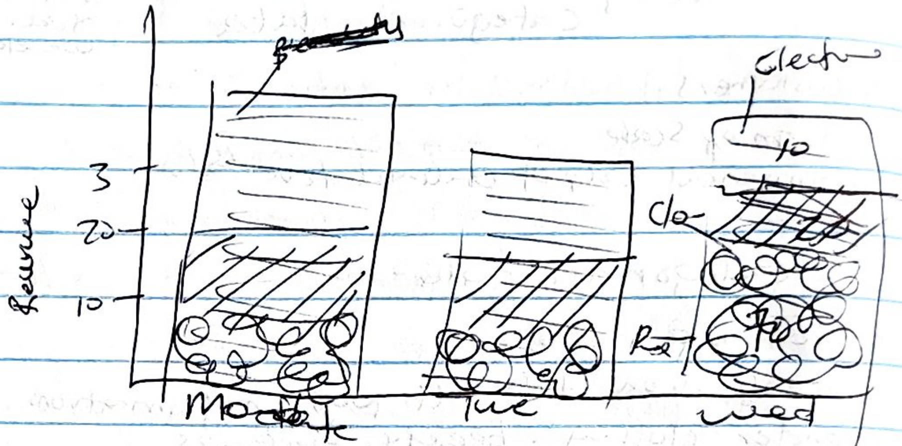
▨ - beauty 60%
▧ - clothing - 20%
▦ - electronics - 20%

100%

②

(Stacked Chart)
Steck-bar graph





You are showing propotion for 2 different categorical data by a spefic metric.

Product category
- beauty and clothing
- Electronics

(Day of week. Category)
Monday
Tue.
Wednesday

EX! 1 Helping the ECO make business decision based on the generation events! Analyse, Present and data visualisation and SQL

1. Excel:

date	Day-name	Day of month	day-clisification	time baskets	Number of Sales	Revenue per day
					2	12
23-01-01	Sunday	Jan	Weekend	01. Morning	4	18.6
23-01-01	Sunday	Jan	Weekend	01. Morning	2	13.5
23-01-01	Sunday	Jan	Weekend	0. Morning	2	4

https:/brav-business-view.loveable.app-

scatter charts".
power BI
presentaton
- Coding-
(Notebook ML
Creating vedios)

Link to dashboard.

-> can download as a fire PDF.

-> What has to be concluded?
- does it make sense
- What Questions did it answers
- What type dool you use

(Loveable)

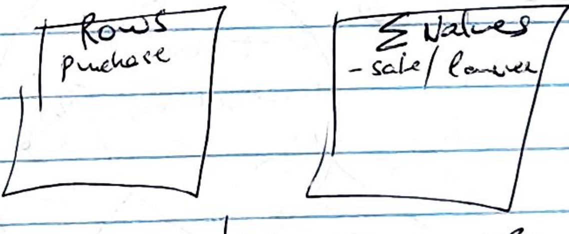
take data to chatGPT
use this data to create a prompt that I can paste and to loveable for creat for me an interactive dashbord and under each graph it should give some insight on what the insight graph shows.
- loggin into loveable

(low code) You can process data and also databricks
Quality of pront can give quality dashbord.

Step 1. top left Δ betwen 1 & A
- Insert -> Perviot atable (Pivotable
- Insert on new spreadshet.

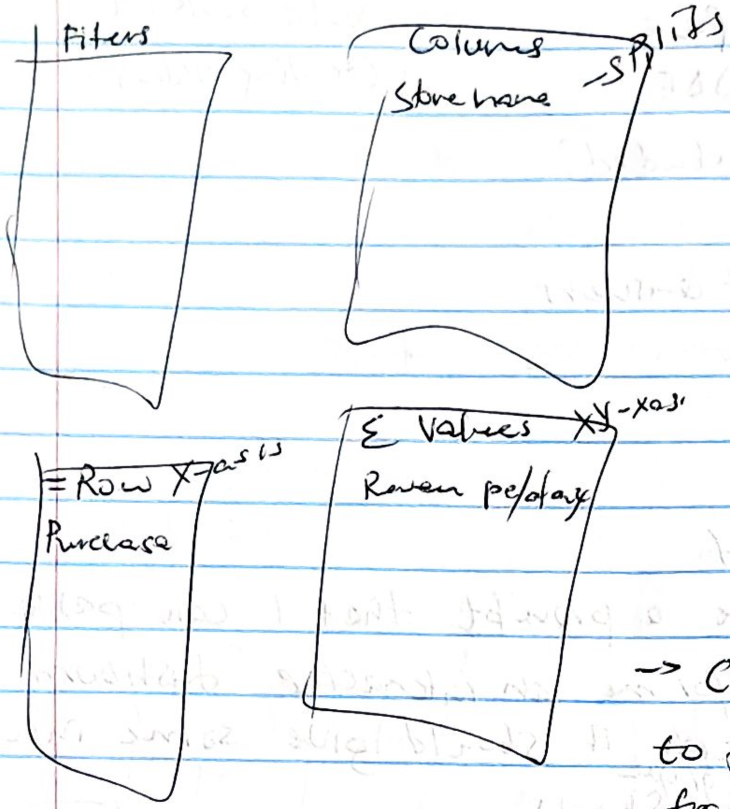
(On the fieldname)
<- Purchace date

> selecting the needed rows and colums paste and drop.

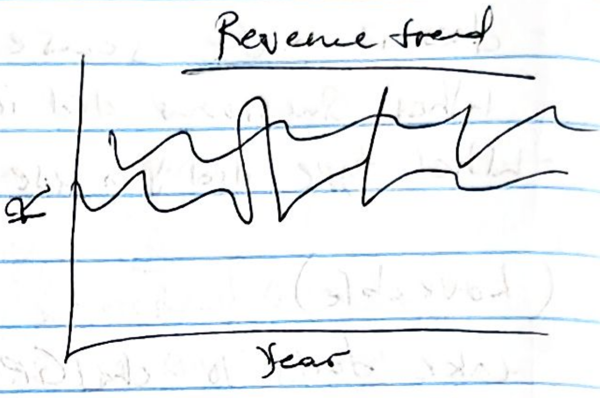


Number of Products.	number of Stores	Spend bucket.	Store Location	Product Category	Product details
1	1	Low Spend	Lower Manhattan	Coffee	Ethopia Rg
1	1	Low Spend	Lower Manhattan	Tea	Spicy Eye Opener
1	1	Low Spend	"	drinking Chocolate	Dark chocolate Lg
1	1	Low Spend	"	Coffee	Our Old time Diner Bill

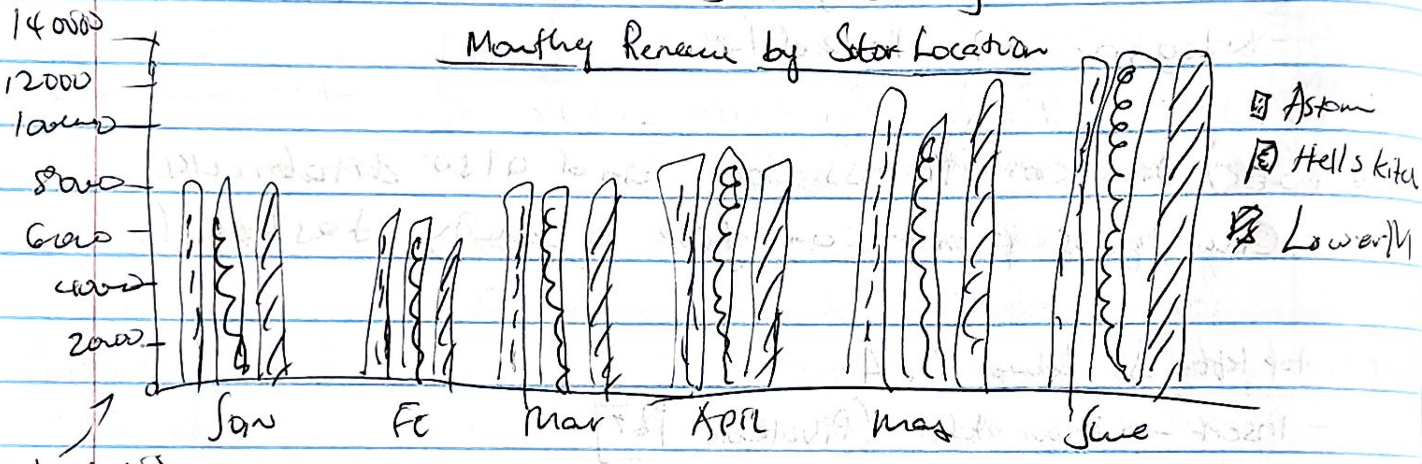
drop and Paste ino Creating new speedsheed;


Filters
Columns
Store name
-> splits
Σ Values
Y-xasis
Reven pe/day
=Row X-asis
Purchase

- Select date\Year
- insert
- select desiog grah

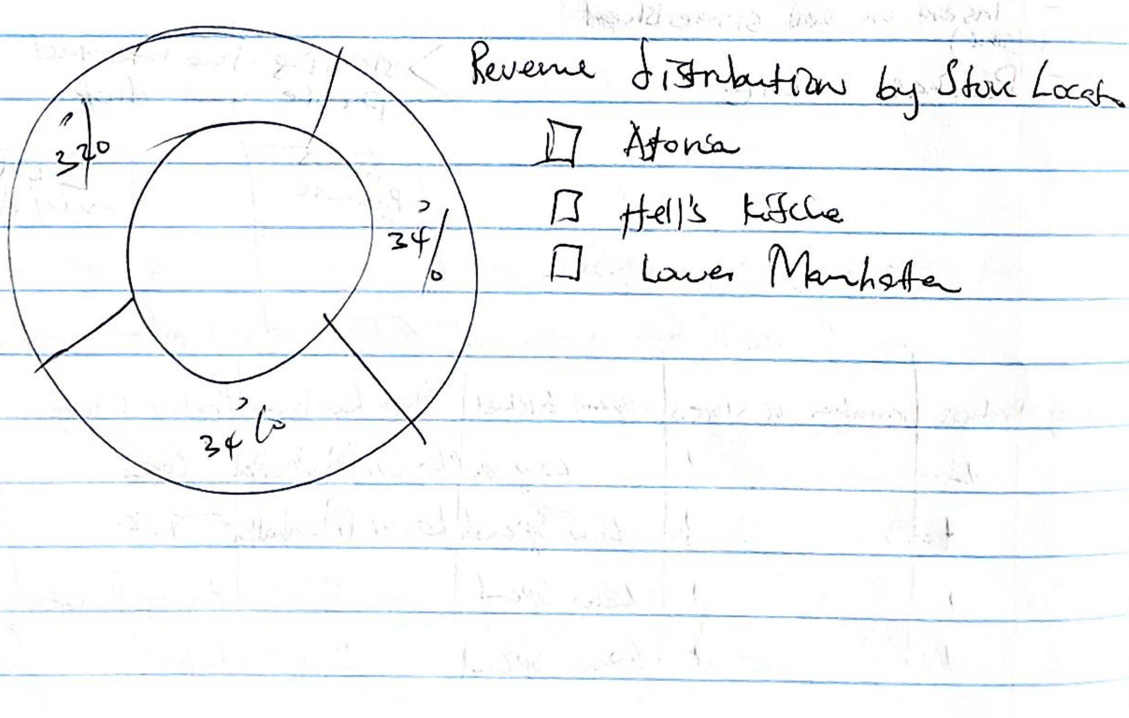


-> Created Pivot-table from Excel to present the renewe trend from coding



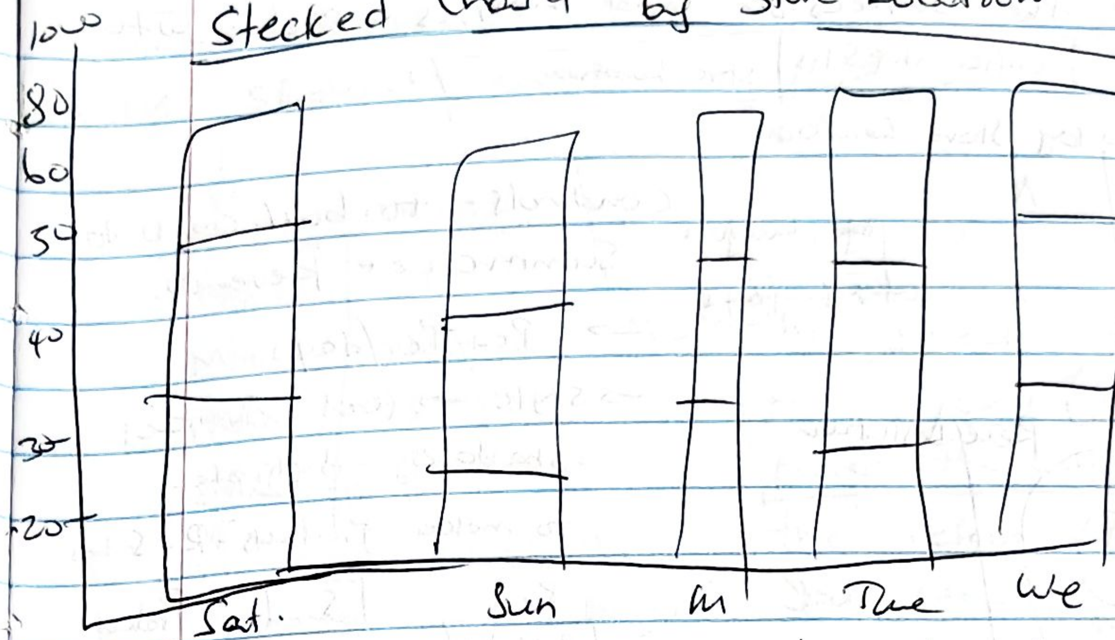
Analysis

Propotions ->



Weekly

Stecked Chard' by Store Locofrow



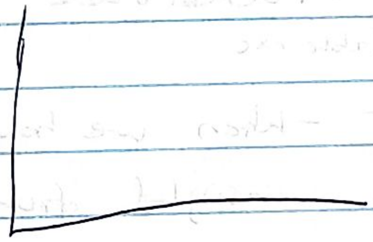
You can also do it IN percentages in the metrics
(bulling data Agents

How? video?

Jafo Studio - Google

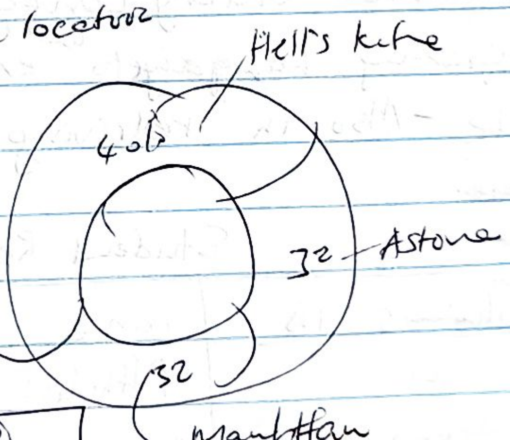
Creat button - Report
CSV File Upload -> creat datset (NAME it) coffee bo1.
-> search for coffee Slop Analysis)
- Add chart ->(Nane a deshborel
↳ Choose Bar - Vertical

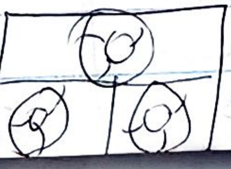
has 2 parts
propeties - Charts
- data



Set Style
choose your metric & dimension - category
↓ # of Sales eg Revenue eg store location

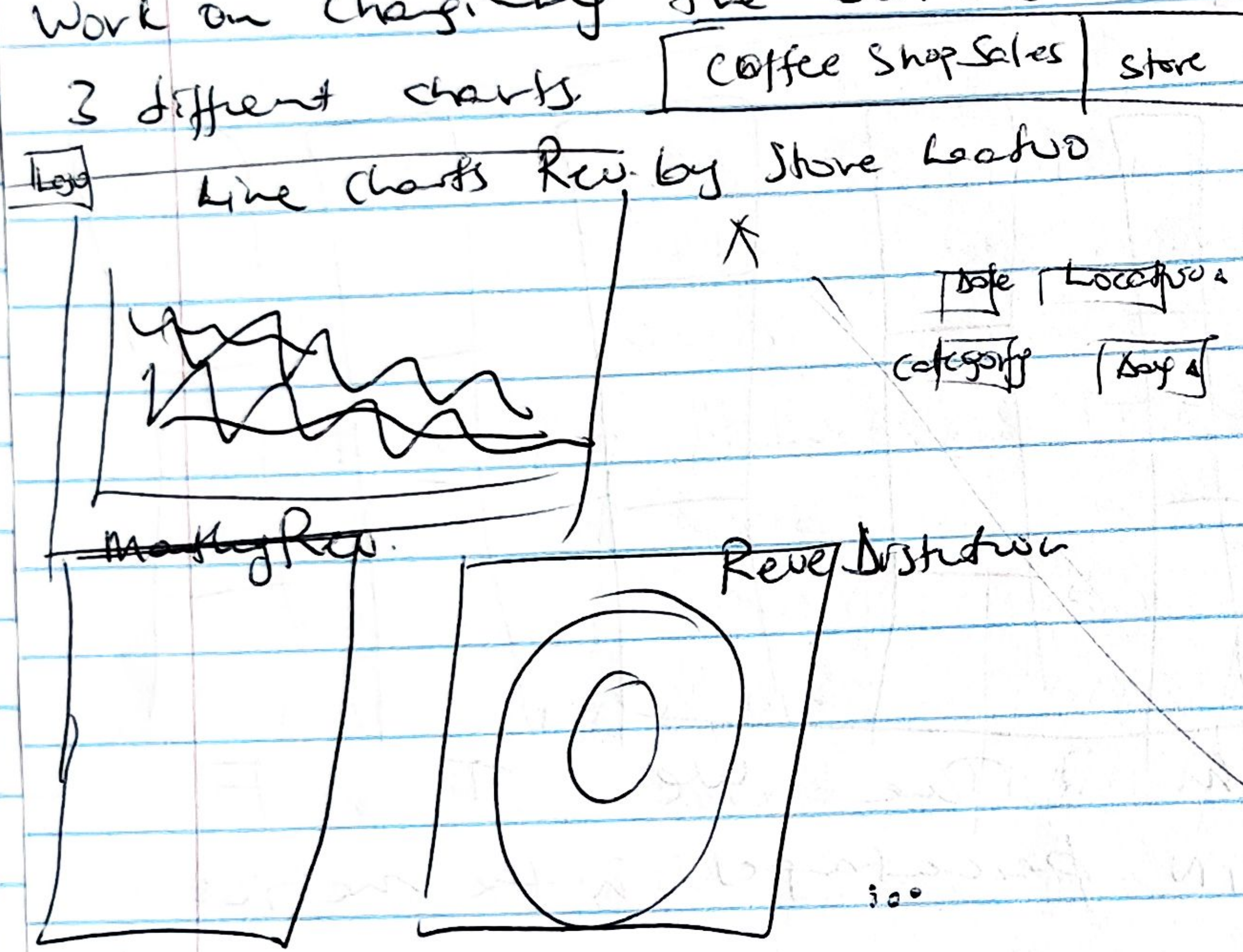
Include chart title
e.g Rev. distribution by Store Location.
(edits) size, style, font, colour
• make all 3 charts allign. then Select different Charts.





All R → Then—

Work on changing the dimension and metrics. ending with the 3 different charts.


Coffee Shop Sales
Store Location
Logo
Line Charts Rev. by Store Locatio
~~Monthly Rev.~~
Reve Distribution
Date
Location
Category
Day
100

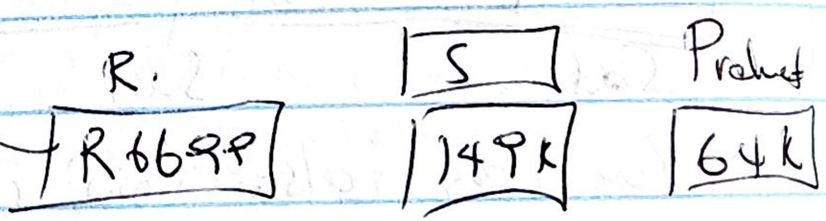
Controls - to built controls to Summarize Revenues.

-> Rev Per/day

-> style -> font -> currency

-shadows - duplicate -

to make Products - R - Sales


R.
S
Product
R6699
149K
64K

-> themes and layout

->Add logo - (from CONA)